Who Guards the Update? Transactional Controller Publication for Physical-World AI

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Changing physical conditions can motivate a better controller, but its installation can still be unreliable. A predecessor invocation may remain live after a candidate becomes current, and a stale request may target a superseded installation. We define a conditional installation contract for one executor slot: check the expected installation, admit stateless candidates, revalidate authority, exclude active invocations, and preserve accounting on rejection. We compare quiescence-free atomic publication (QF-AP) with guarded replacement (GR). Across 200 primary schedules per protocol, QF-AP produced retired predecessor commands in 72 schedules and GR in none. In a separate 50-schedule above-period stress stratum per protocol, QF-AP produced old/new invocation overlap in every schedule; GR prevented overlap but skipped service in every schedule. A corrective replay exposes the counter-hazard: QF-AP publishes immediately but permits one retired command; GR prevents that command but delays publication by 1.137 ms. Here "transactional" refers only to software-authority publication; it implies neither crash atomicity, persistent recovery, nor reversible physical effects. The experiments use hosted guards and simulated control, without physical actuation, foundation-model adaptation, privileged controller execution, or downstream command fencing.

1 Controller updates under changing physical conditions

A payload change can motivate controller retuning. Improved tracking does not establish whether a predecessor invocation can still issue a command after replacement, or whether a delayed request names the current installation. This is a deployment question even when the candidate itself is useful.

Adaptation should not imply deployment.

Physical-world agents can revise skills or controllers as operating conditions change. Executable skill libraries and failure-driven robotic runtimes make such revisions recurrent [Wang et al., 2023, Tian et al., 2026]. Proposals may arrive asynchronously, and concurrent tools need not share a control-cycle boundary. The contract also applies to human-authored updates; repeated adaptation makes installation correctness and update delay part of the evaluation loop.

We contribute a five-part executor contract, a one-slot implementation, and a matched comparison of *quiescence-free atomic publication* (QF-AP) with *guarded replacement* (GR). A payload-shift schedule sweep, hosted timing, and corrective-stop replay separate tracking quality, installation correctness, and availability cost. We claim no new pointer-exchange, version-tag, quiescence, or reference-monitor primitive.

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2 The installation contract

An installation is $I = (h, e)$: a program handle and an epoch that increases on every successful installation, including rollback. Exhaustion rejects instead of wrapping. Bootstrap establishes a slot and authority ceiling; replacement requires an incumbent identity. A Rust research kernel implements the contract at one opt-in TIMER slot. Ordinary fanout and downstream commands are outside this boundary.

C1: Identity and freshness. Compare the full expected (h, e), preventing a delayed A–B–A request from succeeding merely because handle A is current again.

C2: Invocation quiescence. A successful guarded replacement excludes overlap between guarded predecessor and successor invocations. An atomic open/active/transitioning gate admits either invocation or replacement. An active invocation causes replacement to return `Busy`; it does not wait for that invocation.

C3: State admissibility. Admit only candidates with no verifier-reported runtime-map references (`Stateless`). State transfer is unsupported; external and device state are unaffected.

C4: Authority and resource revalidation. Recheck ownership, retained provenance, authority ceiling, verifier constraints, conflicts, stop state, and capacity at installation. Immutable decoded instructions retain load-time metadata; replacement re-verifies them without rechecking the original artifact signature.

C5: Publication and accounting consistency. Runtime and full installation identity share one immutable published object. Admission accounting and the receipt complete within the same serialized manager operation. Every returned error preserves predecessor authority and accounting. These fields are not one multiword atomic snapshot; physical effects and crash recovery are excluded.

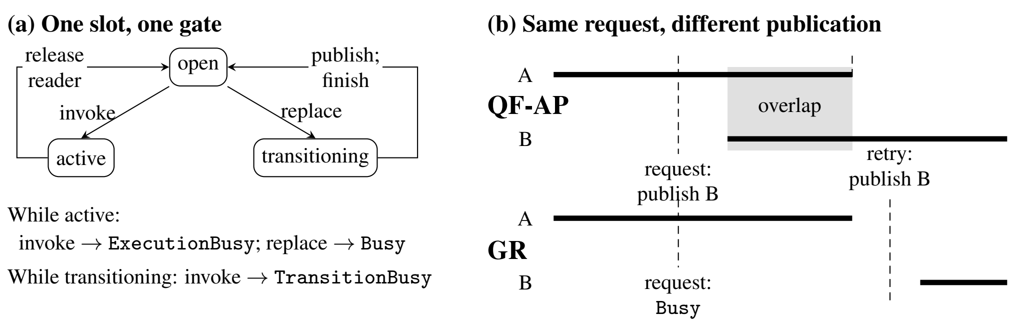

Figure 1: Bars show invocation lifetimes; shading marks overlap; dashed lines mark replacement attempts. Time runs left to right. QF-AP waits for A before returning. The depicted hold crosses a successor dispatch; not every sweep schedule overlaps.

Two invariants. Under serialized manager operations and normal completion, successful open-to-active and open-to-transitioning claims exclude one another. Rust RAII retains the invocation guard through the callback and drops its snapshot pin before reopening. GR publishes while transitioning and reopens afterward; thus predecessor and successor guarded invocations cannot overlap. Invocation linearizes at its successful gate claim; publication linearizes at the complete-object pointer swap.

Returned-error preservation follows from validation, allocation, epoch checks, and final stop checks preceding ledger mutation. Ledger commit validates before writing; subsequent publication and receipt updates have no error return. Diagnostic counters may change. Panic, power failure, unsafe bypasses, and queued commands are outside both arguments.

3 Matched evidence and availability costs

Comparison and measurements. QF-AP and GR use the same manager checks, candidates, accounting, snapshot preparation, and receipts. The host-only QF-AP path omits the invocation gate. It publishes before draining old readers, so publication and API return are different events. GR excludes active readers before publication. Hosted callbacks retain real guards but do not execute privileged bytecode or actuate hardware. Unsigned development fixtures do not test authentication rejection.

Predeclared schedule sweep. Each protocol runs 200 complete eight-second simulations: uniform invocation holds 0/100/500/900 μs crossed with 50 phases spaced 20 μs across a 1 ms dispatch period. A separate 50-phase stratum uses 1,100 μs holds. The plant is $m\dot{v} = u - v$, mass changes from 1 to 2 at 4 s, and the reference alternates ± 1 each second. A feedback tuner produces gains 2/4/8/16 once; both protocols replay those proposals at identical phases. Commands use captured identity and state, complete after the declared hold, and otherwise retain the last output.

Busy requests follow a fixed 137 μs phase walk with a 20 ms request deadline. Completions precede attempts and dispatch; both protocols share an Euler mesh with steps at most 100 μs, including suppressed retry events. No seed or schedule is selected for its outcome. Table 1 reports schedule counts, not failure-probability estimates.

Table 1: Complete replay sweep: 50 schedules per row and protocol, four proposals per schedule. Counts are schedules with at least one event. RMSE is median post-shift tracking error across schedules. The 1,100 μs rows form a separate above-period stress stratum.

Hold (μs)	Protocol	Median post RMSE	Activated/ scheduled	Overlap schedules	Retired-command schedules	Dispatch-skip schedules
0	QF-AP	0.974	200/200	0	0	0
0	GR	0.974	200/200	0	0	0
100	QF-AP	0.974	200/200	0	4	0
100	GR	0.974	200/200	0	0	0
500	QF-AP	0.974	200/200	0	24	0
500	GR	0.974	200/200	0	0	0
900	QF-AP	0.975	200/200	0	44	0
900	GR	0.975	200/200	0	0	0
1100	QF-AP	0.975	200/200	50	50	0
1100	GR	0.976	200/200	0	0	50

Across the primary grid, QF-AP/GR had 0/0 overlap schedules, 72/0 retired-command schedules, and 0/0 dispatch-skip schedules. A retired command can finish after B is published but before B invokes; retirement does not require invocation overlap. At 1,100 μs, GR also serializes same-installation invocations, so task differences include steady-state skipped service, not just the replacement decision. Task quality and installation defects remain separate observations, consistent with governance-oriented evaluation [Qin et al., 2026b].

Hosted logical latency. Ten fresh processes per protocol and hold perform 1,000 measured attempts after 1,000 warmup replacements. Dispatcher and updater use separate physical cores of an AMD Ryzen 7 7735HS, a 1 ms dispatcher period, and identical absolute attempt schedules. A nonallocating clock observation immediately after pointer exchange and reader-epoch flip records publication before reclamation waiting. Clock overhead is retained; this marker follows rather than timestamps the exact atomic instruction.

At the largest primary hold (900 μs), scheduled-request publication median/p99 was QF-AP 6.17/15.00 and GR 6833.08/21887.13 μs. At 1100 μs, GR completed only 3/13 logical requests (10 censored); only 3 processes contributed latency quantiles. Those quantiles exclude unfinished requests; this is a measured progress limitation.

Corrective update. A separate declared replay starts at steady velocity and command 1, mass 1 and reference 2. At 4.250 s it requests gain zero, with a predecessor invocation held from 4.249 to 4.251 s. Both protocols receive the same event. The stop band is $|v| \leq 0.05$ sustained for 100 ms; this is a simulated criterion, not a physical-safety specification.

Over the common fault horizon, QF-AP/GR command integrals were 0.200/2.100 command-ms; retired-command counts were 1/0. Neither protocol is universally safer: early corrective publication competes with predecessor completion and update delay. Quiescence does not fence commands that escape the invocation boundary.

Table 2: Hosted latency in μs: medians across process-level medians/p99s, conditional on completion. Publication columns start at the first scheduled request and include scheduler delay, updater backlog, pre-call setup, Busy attempts and retry gaps through the post-swap observation. Counts include censored requests. Appendix D reports both endpoints, maxima, p95s, threshold denominators and unfinished requests. No real-time guarantees.

Hold (μs)	QF-AP publication median/p99 (μs)	GR publication median/p99 (μs)	GR completed/ started
0	8.52/15.62	8.49/15.81	9992/9992
10	8.57/13.65	8.51/1144.07	9895/9895
100	8.51/13.07	8.76/1147.21	8976/8977
500	7.98/14.52	9.06/4560.70	4950/4955
900	6.17/15.00	6833.08/21887.13	984/993
1100	12410.30/29799.71	334943.52/334943.52	3/13

Table 3: Corrective replay. Delay is from the common fault/request to observed publication; command integral uses the common fault-to-8 s horizon. Stop time is band entry after the fault, confirmed for 100 ms. Command values are simulation units.

Protocol	Publication delay (μs)	Stop entry after fault (ms)	$\int_{fault} \lvert u \rvert dt$ (command ms)	Retired commands	Skips/ retries
QF-AP	0	2995.800	0.200	1	0/0
GR	1137	2997.700	2.100	0	1/1

Supporting checks. Both protocols preserve six rejecting fault requests and reject stale epochs; concurrent proposers have one winner. Attach-first/detach-first remain interruption witnesses. Shared-source Loom models slot entry/transition races and snapshot reclamation, not the complete manager or plant. A separate QEMU probe exercises interpreter A–B–A dispatch. Appendix C separates these scopes.

4 Established mechanisms and implications

ROS 2 lifecycle states govern managed processing, but the design does not specify a concurrent callback-drain and conditional-version transaction [Macenski et al., 2022, Biggs and Foote, 2021]. Kitsune uses explicit update points and state transformation; Linux livepatch moves tasks individually when safe [Hayden et al., 2014, Linux Kernel documentation contributors, n.d.a]. RCU permits old readers after new publication [Linux Kernel documentation contributors, n.d.b]. Linearization points and ABA defenses are established concurrency concepts [Herlihy and Wing, 1990, Michael, 2004].

Qin et al. govern compatibility, staged evaluation, activation, and rollback; their inspected procedures do not specify our exact-incumbent operation with invocation exclusion and ledger consistency [Qin et al., 2026a]. LITHE obtains quiescence at a single Spine-cycle boundary and retains shared controller state [Lim and Clites, 2026]; a concurrent executor need not have that boundary. Action shielding and Neural Simplex address complementary behaviour-level assurance [Alshiekh et al., 2018, Phan et al., 2020].

GR does not wait for active invocations, but it is not inherently starvation-free: progress requires the updater to claim an open interval. Delaying a corrective update is a counter-hazard. The deployment tradeoff concerns latency, overlap and service; it is not the classical parameter-learning stability–plasticity tradeoff. Stateful updates, downstream freshness, persistent recovery, physical safety and foundation-model learning remain unevaluated.

Physical-world AI evaluations should report task quality and installation correctness separately, including update latency, stale-version execution, retries, skipped service, and recovery outcomes.

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A Protocol API and state machine

Installation identity is (h, e), distinct from the two reclamation-reader epochs. An installation epoch increases after every success, including rollback, and never wraps. Bootstrap establishes the authority ceiling. A candidate is a loaded resident program. Original signed artifact bytes are not retained: the runtime stores decoded instructions and load-time metadata, then rechecks verification and authorization at installation.

```
invoke(callback):
  claim open→active or return ExecutionBusy/TransitionBusy
  read immutable (runtime, identity); if empty, reopen and reject
  call callback while retaining the invocation guard
  drop snapshot pin; reopen; return callback result
try_replace(owner, expected, candidate, Stateless):
  validate candidate and absence of referenced maps
  claim open→transitioning or return Busy
  check expected identity, owner, provenance, authority and conflicts
  reverify TIMER constraints; preflight accounting delta and next epoch
  prepare snapshot; acquire final clear-stop gate or reject
  validate and commit accounting delta
  exchange immutable (runtime, identity) pointer [linearization]
  finish publication; set manager epoch, identity and receipt
  release stop gate; reopen; return receipt
```

All error exits release any acquired transition guard. Candidate checks precede gate acquisition, so an invalid candidate can return its specific error while another invocation is active. Receipt fields identify slot, predecessor and installed identity. Same-handle rollback is a new installation rather than reversal of time.

QF-AP uses the same commit preparation and accounting. Its diagnostic publication omits the invocation gate: after pointer exchange and reader-epoch flip it waits for the previous reader counter, then reclaims. During that wait, a reader can observe B while manager-protected identity and receipt still describe A; manager access remains serialized and unavailable. Those fields are coherent on return, not co-published. GR has excluded participating predecessor invocations before committing. Production dispatch does not propagate installation identity to downstream actuation.

B Assumptions and proof details

Assume serialized manager operations, valid use of the opted-in slot, guards retained through callback execution, and normal completion after ledger commit. Exclude unsafe bypasses, ordinary TIMER fanout, panic, power loss and commands escaping the invocation. Rejected operations can change diagnostic counters. Bootstrap and deliberate teardown are outside replacement's nonempty-slot invariant.

Invocation exclusivity. Entry and guarded transition claim the same sequentially consistent state from open. Atomic modification order admits at most one. Active excludes publication; transitioning excludes invocation. Rust RAII releases the old epoch reader before reopening, while the transition publishes the successor before reopening. Thus no guarded predecessor invocation overlaps a guarded successor invocation. This is not a command-consumer freshness proof.

Returned-error preservation. Candidate, identity, authority, verifier, quota, epoch and allocation checks precede mutation. Final stop-state and lock-contention rejection precede ledger commit. The ledger revalidates before writing; publication and receipt updates afterward have no error return. Consequently a returned error preserves snapshot, installation identity, ledger and last receipt. This argument promises neither crash rollback nor unchanged concurrent skip counters.

Freshness and accounting. After $A_{e_1} \to B_{e_2} \to A_{e_3}$, the old expected (A, e_1) fails equality with (A, e_3). Epoch exhaustion rejects. Serialized proposers expecting one incumbent have at most one winner; admissibility or a busy gate can leave no winner. Completed-request retries reject rather than act as idempotent acknowledgements. Admission preflight checks $C - c_A + c_B$ before commit. Failed updates leave C unchanged; resident-memory charges for both loaded programs persist independently of the one active-slot charge.

C Model and integration coverage

Loom substitutes instrumented atomics into the real `EpochSnapshot` and `ExclusiveSlot` implementation. Its finite models exercise coherent payload reads, old-reader lifetime and reclamation, competing entry, entry-versus-transition races, and skip counts. It does not model the complete manager, authority verifier, admission ledger, receipt update, plant, or diagnostic QF-AP bypass. Model-check success is not unrestricted whole-kernel verification.

The hosted manager test and paired campaign separately check conditional activation, rejection preservation, accounting, rollback, ABA and competing proposers. Callbacks retain real guards but perform host arithmetic rather than privileged BPF execution. The QEMU probe checks interpreter dispatch after A–B–A and stale-request rejection; it does not validate physical hardware or controller safety. In the following tables P means preserved snapshot, full identity, admission ledger and receipt.

Table 4: Paired campaign outcomes. Error names are implementation outcomes; P denotes predecessor preservation. The successful rollback and concurrent/held-reader schedules are separate from the six rejecting requests per protocol.

Schedule	QF-AP	GR
Snapshot preparation	`SnapshotAllocationFailed`; P	Same
Over-budget	`AdmissionRejected`; P	Same
Authority expansion	`AuthorityExceeded`; P	Same
Stale expected identity	`StaleInstallation`; P	Same
Completed-request retry	`StaleInstallation`; P	Same; no recharge
Rollback	`Ok`; A with fresh epoch	Same
ABA stale epoch	`StaleInstallation`; P	Same
Concurrent proposers	One `Ok`, one `StaleInstallation`	Same
Held predecessor	`Ok`; B current while A live	`Busy`; P; retry `Ok`

Table 5: Additional retained integration-test coverage. The log records the enclosing test passing; individual assertions are inspectable in its source, not separate paired trace rows. "Not observed" must not be read as a passing QF-AP result.

Condition	GR assertion	QF-AP evidence
State transfer	`StateTransferUnsupported`	Same error asserted
Map-bearing candidate	`PersistentStateUnsupported`; P	Same error and P asserted
Epoch exhausted	`EpochExhausted`; P	Not observed; shared check
Stop gate held/active	`EmergencyStopBusy`; `EmergencyStopActive`; P	Not observed; shared commit
Active-owner teardown	Reclamation returns false; identity preserved	Active case not observed

Successful teardown intentionally clears the slot; it is not a replacement that preserves nonempty publication. The separate QF-AP teardown success follows explicit slot clearing and does not establish behavior during an active QF-AP invocation. Authentication rejection is not exercised by these unsigned-development fixtures. Attach-first and detach-first provide supporting interruption witnesses, not equivalents to the same-check QF-AP baseline. Loom checks the gate and snapshot machinery independently; finite model checks and fault schedules do not establish unrestricted whole-kernel correctness.

D Hosted measurement and complete distributions

Both protocols use the same preloaded candidates, 1 ms dispatcher period and absolute updater phases. Ten fresh processes per hold use 1,000 excluded warmup operations followed by 1,000 measured attempts. Holds are imposed guard durations, not bytecode runtimes. Dispatcher CPU 12 and updater CPU 14 are on separate physical cores of the retained Ryzen 7 7735HS; clock resolution, compiler/release flags, affinity, governor and boost state are recorded in the environment manifest. OS lateness, missed releases and tails remain in the data. The hosted dispatcher executes late releases immediately while lateness is below one period, and expires older releases. Long callbacks can therefore cause near-continuous catch-up occupancy. The replay instead attempts each fixed tick and GR skips active ticks. Their progress results describe different declared arrival policies, not a universal duty-cycle threshold.

Attempt j in process r is scheduled at $t_0 + j\,\mathrm{ms} + ((137j + 97r) \bmod 1000)\,\mu\mathrm{s}$. Busy retains the same logical request; success advances its expected identity and candidate. The clock is `CLOCK_MONOTONIC_RAW`. The timed callback stores only the post-swap/epoch-flip clock observation in preallocated storage, before reclamation waiting; no extra observer reader, allocation or logging is inserted. Clock failure invalidates capture without an error return after commit.

For first scheduled release s, first API-call entry a (the called timestamp), successful post-swap observation p, and API return r, derive all four intervals $p - s, p - a, r - s, r - a$. Publication and return must not be substituted. Quantiles use nearest-rank $\lceil qn \rceil$. Aggregate latency cells use median of process quantiles, and the largest observed maximum. They condition on completion rather than assigning a latency to unfinished requests. The started timestamp precedes pre-call setup; scheduled intervals include that setup and initial scheduler delay. Table 2 uses the first scheduled request as its origin. Table 6 reports first-API-call-origin intervals, while Table 7 uses the scheduled origin for threshold accounting. All four endpoints remain in derived data. Processes with no completion contribute counts but no latency quantile. The hosted campaign has an attempt budget, not a 20 ms retry deadline; that deadline belongs to the separate replay.

Table 6: Hosted first-attempt latency distributions in μs. Each quartet is median/p95/p99/maximum, with the aggregation above. Counts are completed/started logical requests; unfinished requests remain censored.

Hold (μs)	Protocol	First call to publication med/p95/p99/max (μs)	First call to return med/p95/p99/max (μs)	Completed/ started
0	QF-AP	1.90/3.40/6.30/65.43	1.94/3.47/6.50/65.78	10000/10000
0	GR	1.89/3.39/6.26/1142.85	1.93/3.44/6.40/1142.90	9992/9992
10	QF-AP	1.92/3.30/5.60/51.33	1.97/3.60/7.62/51.48	10000/10000
10	GR	1.90/3.67/1138.48/1167.62	1.94/3.76/1138.53/1167.73	9895/9895
100	QF-AP	1.91/3.22/5.14/32.49	1.98/53.72/93.56/103.15	10000/10000
100	GR	1.92/1140.64/ 1142.08/2278.96	1.98/1140.69/ 1142.12/2279.00	8976/8977
500	QF-AP	1.86/3.49/5.70/98.23	7.95/458.99/499.53/1310.68	10000/10000
500	GR	2.02/4552.20/ 4555.94/5707.49	2.07/4552.24/ 4556.09/5707.92	4950/4955
900	QF-AP	1.33/3.21/4.79/38.01	402.49/857.01/ 901.15/1413.10	10000/10000
900	GR	6827.07/14926.37/ 21880.68/65913.02	6827.18/14926.44/ 21880.75/65913.13	984/993
1100	QF-AP	1.11/2.83/5.12/42.42	1105.54/1106.88/ 1109.40/1635.85	10000/10000
1100	GR	334935.09/334935.09/ 334935.09/552804.55	334935.49/334935.49/ 334935.49/552804.91	3/13

Raw schema 3 records full installation identity before and after every timed call, sampled outside its latency interval. Busy must preserve identity; successful epochs advance once. An earlier timing-only capture is retained separately and is not pooled: recapture added these observed identities, without selecting schedules or outcomes. The two complete replay captures are byte-identical; the retained package stores one copy.

At 0 μs hold, successful observer round-trip was 181 ns (run-median range 180–181 ns) for QF-AP and 200 ns (run-median range 200–220 ns) for GR. At 500 μs hold, GR Busy return latency was 70/270 ns median/p99; the separately controlled TransitionBusy return was 60/90 ns median/p99. The manager-accounted resident bytecode was 16 bytes for one loaded candidate and 32 bytes for two; these counters exclude allocator, metadata, and process RSS.

Table 7: Scheduled-request publication threshold accounting, summed across processes. Each threshold cell is known met / known missed / unknown; its denominator is every logical request started. A censored request observed for less than the threshold remains unknown.

Hold (μs)	Protocol	1ms met/missed /unknown	5ms met/missed /unknown	10ms met/missed /unknown	20ms met/missed /unknown
0	QF-AP	9998/2/0	10000/0/0	10000/0/0	10000/0/0
0	GR	9986/6/0	9992/0/0	9992/0/0	9992/0/0
10	QF-AP	9998/2/0	10000/0/0	10000/0/0	10000/0/0
10	GR	9798/97/0	9895/0/0	9895/0/0	9895/0/0
100	QF-AP	10000/0/0	10000/0/0	10000/0/0	10000/0/0
100	GR	7969/1007/1	8976/0/1	8976/0/1	8976/0/1
500	QF-AP	9998/2/0	10000/0/0	10000/0/0	10000/0/0
500	GR	3577/1377/1	4945/5/5	4950/0/5	4950/0/5
900	QF-AP	10000/0/0	10000/0/0	10000/0/0	10000/0/0
900	GR	69/923/1	73/915/5	584/401/8	950/34/9
1100	QF-AP	412/9588/0	2454/7546/0	4647/5353/0	6810/3190/0
1100	GR	1/12/0	1/12/0	1/12/0	1/12/0

The raw and derived records retain per-request attempt counts, Busy counts, scheduled/actual times, per-process quantiles, dispatcher skips, resource counts and lateness. Rates with no conditioning population are undefined. Skip-per-activation divides aggregate natural skips by successful activations; forced transition probes are separate. This campaign estimates hosted costs under a declared schedule, not worst-case time or eventual progress.

E Complete control-replay specification

The primary grid has exactly four holds times 50 phases times two protocols. The separate stress stratum adds 50 phases per protocol at 1,100 μs. Each case has a fresh manager, identical resident candidates and an eight-second plant trajectory. Initial velocity and held command are zero; mass is 1 before 4 s and 2 afterward; $r(t) = (-1)^{\lfloor t/1\mathrm{s} \rfloor}$. Commands are $u = \mathrm{clip}(k(r - v), -4, 4)$ and the Euler plant step is

$$v(t + \Delta) = v(t) + \Delta\,(u_{\mathrm{held}} - v(t))/m(t), \quad \Delta \leq 100\,\mu\mathrm{s}.$$

The mesh is fixed for the matched protocols from the regular integration grid and every potential attempt and completion time, including suppressed attempts. Protocol-dependent extra Euler subdivisions must not affect task comparisons.

The seed tuner uses preceding 250-tick RMSE, threshold 0.1, initial gain 1 and cap 16, at 4.25/4.5/4.75/5 s. Its frozen gains are 2/4/8/16. Grid primaries occur at those bases plus phase p. Retry j occurs at $b + j \cdot 1000 + ((p + 137j) \bmod 1000)\,\mu$s, only following Busy and strictly before primary+20 ms. Expired requests are retained. No completed request receives a synthetic retry.

At equal timestamps finish invocations in start order, apply their captured commands, finalize completed QF-AP calls, process attempts, then dispatch. Zero-hold invocation completes immediately. Commands capture identity, reference, velocity and gain at entry; later completion applies the captured value. Skipped dispatch retains output. Observing QF-AP's published identity does not acquire the manager lock held by its pending writer. Cleanup drains all workers, clears the exact final installation, unloads resident handles and checks accounting.

RMSE uses 4,000 ticks before and 4,000 after the mass change, including skipped ticks. Record maximum absolute error/command, saturation at the declared bound and violations above 4 separately. These are simulation envelopes, not physical constraints. Scenario fractions describe a census of the declared grid. Per-request retry/deadline populations differ from the 50-scenario populations; complete CSV rows retain both denominators.

The legacy replay retains its one special 2 ms hold and paired synchronous 250 ms utility probes as a separate witness. It is not pooled into the primary grid. No random seed or post-hoc choice is involved.

Corrective case. Use mass 1, $v(0) = 1$, held command 1, reference 2, gain 1 and a single gain-zero proposal at 4.250 s. Normal holds are 100 μs; one invocation starting at 4.249 s ends at 4.251 s. The same retry policy applies. Integrate absolute command over both the common fault-to-8 s horizon and each publication-to-8 s horizon; report the latter as different windows. Stop-band entry requires $|v| \leq 0.05$ continuously for 100 ms, with entry and confirmation recorded or censored at the horizon. Post-fault tracking error retains reference 2; the stop criterion instead targets zero velocity. A retired completion is an installation defect even when its task consequence is small or absent.

F Conditional comparison with established mechanisms

ROS 2 lifecycle exposes supervised states and managed-processing policies; its design does not specify a linearized drain of in-flight callbacks [Biggs and Foote, 2021]. Kitsune obtains all-thread quiescence at explicit update points and transforms state. Livepatch instead converges tasks individually and may remain transitional indefinitely [Hayden et al., 2014, Linux Kernel documentation contributors, n.d.a]. RCU deliberately permits old readers after new publication [Linux Kernel documentation contributors, n.d.b]. These mechanisms supply different consistency boundaries rather than interchangeable safety guarantees.

Table 8: Conditional mechanism comparison. Full and partial refer only to the stated boundary; "not specified" is not a claim of impossibility. The source-by-source literature notes retain broader comparisons.

Obligation	Established support	This executor boundary
Identity	Version registries: partial; tagged identity addresses ABA	Full expected (h, e); no wrapping
Quiescence	DSU barriers: full at update points; RCU: permits coexistence	GR full for guarded invocations; QF-AP omits it
State	Kitsune transforms state; LITHE retains shared state	Stateless only; transfer unsupported
Authority	Qin checks policy/context; ROS supplies supervision	Revalidate authority at installation
Accounting	Qin checks declared resources; ledger commit not specified	Serialized delta and receipt; runtime/id co-published
Downstream freshness	Action assurance constrains behaviour	Not provided by this prototype
Progress	DSU convergence depends on safe points	Open interval required; no fairness claim
Retry/skip outcomes	ROS/LITHE expose failures or skipped work	Explicit observed counts and deadlines

Qin's lifecycle governs whether a version should advance; the exact-incumbent concurrency and accounting protocol is not specified in the procedures inspected [Qin et al., 2026a]. LITHE's single Spine cycle supplies controller quiescence without establishing downstream command fencing [Lim and Clites, 2026]. Our identity check addresses the established ABA problem; the publication point uses linearizability [Michael, 2004, Herlihy and Wing, 1990]. The contribution is the combined, tested contract and its availability consequences, not invention of those primitives.

G Reproducibility and evidence availability

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